

Full Stack Development Courses (Java)

Frontend Technologies:

1) HTML

- **Introduction to HTML**
- **HTML Document Structure** (doctype, head, body, meta tags)
- **Text Formatting & Semantic Tags** (p, h1-h6, strong, em, etc.)
- **Lists & Tables** (ul, ol, dl, table, tr, th, td)
- **Forms & Input Elements** (input, textarea, select, button, label)
- **Links & Navigation** (a, nav, href, target, anchor links)
- **Images & Multimedia** (img, audio, video, iframe)
- **HTML5 New Elements** (article, section, aside, header, footer)
- **Forms & Validation** (required, pattern, minlength, maxlength)
- **CSS Integration** (inline, internal, external CSS)
- **HTML APIs** (Geolocation, Drag & Drop, Web Storage)
- **Responsive Design Basics** (meta viewport, media queries)
- **SEO Best Practices** (meta tags, alt attributes, structured data)
- **HTML Accessibility** (ARIA attributes, semantic elements)
- **Basics of JavaScript with HTML** (script tag, event handling)
- **Client Server Architecture:**

2) CSS

- **CSS Syntax & Selectors** (class, id, universal, attribute, pseudo-classes)
- **Box Model** (margin, border, padding, content)
- **Flexbox** (display: flex, justify-content, align-items, flex-wrap)
- **Grid Layout** (grid-template-rows, grid-template-columns, gap)
- **Positioning** (static, relative, absolute, fixed, sticky)
- **Responsive Design** (media queries, viewport units, rem/em)
- **Typography** (font-family, font-size, line-height, text-align)
- **CSS Transitions & Animations** (transition, @keyframes, animation)
- **CSS Variables** (custom properties, var())
- **Backgrounds & Borders** (background-image, gradients, border-radius)
- **Pseudo-elements** (::before, ::after)

- Z-index & Stacking Context
- CSS Filters & Effects (blur, grayscale, drop-shadow)
- CSS Frameworks Basics (Bootstrap, Tailwind CSS)
- Dark Mode & Theming (prefers-color-scheme, CSS variables)

3)JavaScripts

- JavaScript Basics (Variables, Data Types, Operators)
- Control Flow (if-else, switch, loops)
- Functions (Function Declaration, Expression, Arrow Functions, Callback Functions)
- ES6+ Features (let/const, Template Literals, Destructuring, Spread/Rest Operators)
- Objects & Arrays (Properties, Methods, Object Destructuring, Array Methods like map, filter, reduce)
- DOM Manipulation (getElementById, querySelector, Event Listeners, Modifying Elements)
- Events & Event Handling (onClick, onChange, addEventListener)
- Asynchronous JavaScript (Promises, async/await, Callbacks)
- Fetch API & AJAX (GET, POST Requests, Handling Responses)
- Closures & Scope (Global vs. Local Scope, Lexical Scope, Hoisting)
- Prototype & Inheritance (Prototype Chain, Classes, Object-Oriented JS)
- Error Handling (try-catch, throw, finally)
- Modules & Import/Export (ES6 Modules, CommonJS)
- Local Storage & Session Storage
- Event Loop & Call Stack (Execution Context, Callback Queue, Microtasks)
- Regular Expressions (RegExp) (Pattern Matching, Validation)
- Web APIs (Geolocation, Clipboard, Notifications, Speech Recognition)
- Frameworks & Libraries Basics (React.js, Vue.js, Node.js, Express.js)
- Testing & Debugging (console.log, breakpoints, Jest, Mocha)
- Best Practices & Performance Optimization (Debouncing, Throttling, Memory Leaks)

4)Advanced JavaScript

- Execution Context & Call Stack
- Scope & Closures
- Hoisting & Temporal Dead Zone (TDZ)
- Prototype & Prototypal Inheritance

- this Keyword & Binding (call, apply, bind)
- Higher-Order Functions & Functional Programming
- Event Loop, Microtasks & Macrotasks
- Promises, Async/Await & Error Handling
- Debouncing & Throttling
- JavaScript Modules (ES6 Import/Export)
- Generators & Iterators
- WeakMap & WeakSet (Memory Management)
- JavaScript Design Patterns (Singleton, Factory, Observer, etc.)
- Web Workers & Multithreading in JavaScript
- Proxy & Reflect API
- Symbol & Meta-Programming
- Service Workers & Progressive Web Apps (PWA)
- Event Delegation & Bubbling vs. Capturing
- Canvas API & WebGL for Graphics
- Testing & Debugging (Jest, Mocha, Chai, Cypress)

5)Bootstrap

- Introduction to Bootstrap (CDN, Installation, Setup)
- Bootstrap Grid System (Containers, Rows, Columns, Breakpoints)
- Typography & Text Utilities (Headings, Font Styling, Alignment)
- Bootstrap Colors & Backgrounds (Text & Background Utility Classes)
- Forms & Form Controls (Input Fields, Select, Validation, Custom Forms)
- Buttons & Button Groups (Styles, Sizes, Outline Buttons)
- Navbar & Navigation Components (Navbar, Breadcrumb, Pagination)
- Cards & Content Containers (Cards, Jumbotron, List Groups)
- Images & Media (Responsive Images, Figures, Embedding Videos)
- Modals & Popovers (Dialogs, Alerts, Tooltips)
- Bootstrap Components (Accordion, Carousel, Spinners, Progress Bars)
- Tables & Table Styling (Responsive Tables, Striped, Bordered)
- Flexbox Utilities (Alignment, Justify, Ordering)
- Bootstrap Forms Validation
- Grid System & Responsive Design
- Utility Classes (Margins, Padding, Borders, Shadows)
- Bootstrap Icons & Glyphicons
- Dark Mode & Theming with Bootstrap
- Bootstrap with JavaScript (Collapse, Dropdowns, Tabs)
- Customizing Bootstrap (SASS, Variables, Custom Themes)

6) React

- Introduction to React (JSX, Virtual DOM, React vs. Vanilla JS)
- React Components (Functional vs. Class Components)
- Props & State Management
- Event Handling in React
- Conditional Rendering
- Lists & Keys in React
- Hooks in React (useState, useEffect, useRef, useContext, useReducer)
- React Router (Navigation & Routing)
- Context API & State Management
- Forms Handling & Validation
- React Lifecycle Methods (Class Components)
- Higher-Order Components (HOC)
- Controlled vs. Uncontrolled Components
- Error Boundaries & Error Handling
- React Fragments & Portals
- Lazy Loading & Code Splitting (React Suspense & Lazy)
- React Performance Optimization (Memoization, useMemo, useCallback)
- Working with APIs (Fetch, Axios, Async/Await)
- Redux & State Management Libraries (Redux Toolkit, Zustand, Recoil)
- React Testing (Jest, React Testing Library, Cypress)
- Server-Side Rendering (SSR) & Static Site Generation (SSG)
- React with TypeScript
- Next.js (React Framework for SSR & SSG)
- Integrating Third-Party Libraries (Tailwind, Material-UI, Chart.js)
- Progressive Web Apps (PWA) with React

Database Technologies :

1)SQL

- SQL Basics (Data Types, Tables, Databases)
- SELECT Statement & Data Retrieval
- WHERE Clause & Filtering Data

- ORDER BY & Sorting
- DISTINCT & Eliminating Duplicates
- Joins (INNER, LEFT, RIGHT, FULL, SELF, CROSS)
- GROUP BY & HAVING Clause
- Aggregate Functions (COUNT, SUM, AVG, MIN, MAX)
- Subqueries & Nested Queries
- SQL Constraints (PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, CHECK, DEFAULT)
- Indexes & Performance Optimization
- Views (Creating & Managing Views)
- Transactions (COMMIT, ROLLBACK, SAVEPOINT)
- Stored Procedures & Functions
- Triggers & Event Handling
- Window Functions (RANK, DENSE_RANK, LEAD, LAG)
- Common Table Expressions (CTE)
- Recursive Queries
- SQL Security & User Management
- SQL Injection Prevention

2)MongoDB

- Introduction to MongoDB (NoSQL vs. SQL, Features, Use Cases)
- MongoDB Installation & Setup
- MongoDB Data Model (Documents, Collections, Databases)
- CRUD Operations (Create, Read, Update, Delete)
- MongoDB Query Language (MQL)
- Indexes in MongoDB (Single Field, Compound, Multikey, Text, Geospatial)
- Aggregation Framework (Pipeline, \$match, \$group, \$sort, \$lookup)
- MongoDB Relationships (Embedded Documents, Referenced Documents)
- Schema Design & Data Modeling
- MongoDB Transactions (ACID Compliance, Multi-Document Transactions)
- Replication & High Availability (Replica Sets)
- Sharding & Scalability (Horizontal Scaling, Shard Keys)

- MongoDB Security (Authentication, Authorization, Role-Based Access Control)
- MongoDB Performance Optimization (Indexing, Query Optimization, Caching)
- Working with MongoDB in Different Programming Languages (Node.js, Python, Java)
- MongoDB with Mongoose (for Node.js)
- MongoDB Atlas & Cloud Deployment
- Backup & Restore Strategies
- MongoDB Change Streams & Real-time Data Processing
- Monitoring & Administration (Profiler, Logs, Performance Tuning)

Backend Technologies

1) JAVA

- Core Java Concepts (OOPs, Collections, Exception Handling, Multithreading)
- Java I/O & File Handling
- Java Streams & Lambda Expressions
- JDBC (Java Database Connectivity)
- Servlets & JSP (Java Server Pages)
- Spring Framework (Spring Boot, Spring MVC, Spring Security, Spring Data)
- RESTful API Development with Spring Boot
- Microservices Architecture (Spring Cloud, Service Discovery, Load Balancing)
- Authentication & Authorization (JWT, OAuth2, Spring Security)
- Database Integration (JPA, Hibernate, MySQL, PostgreSQL, MongoDB)
- Caching Mechanisms (Redis, Ehcache, In-Memory Caching)
- Message Queues & Event-Driven Architecture (RabbitMQ, Kafka)
- Unit Testing & Integration Testing (JUnit, Mockito, TestContainers)
- Asynchronous Processing & Multithreading (CompletableFuture, ExecutorService)
- Logging & Monitoring (Log4j, SLF4J, ELK Stack, Prometheus)

- Containerization & Deployment (Docker, Kubernetes, CI/CD Pipelines)
- GraphQL vs. REST API (Using GraphQL in Java)
- WebSockets & Real-time Communication
- Serverless Backend with Java (AWS Lambda, Azure Functions)
- Performance Optimization & Scalability

2)Spring Boot 3

- Introduction to Spring Boot 3
- Spring Boot Starter Dependencies
- Auto-Configuration & Spring Boot Properties
- Spring Boot Annotations
- YAML vs. Properties File Configuration
- Spring MVC & RESTful API Development
- Exception Handling in Spring Boot
- Validations with Spring Boot
- Spring Boot with Databases (JPA, Hibernate, JDBC)
- Spring Data JPA & Repository Pattern
- Working with PostgreSQL, MySQL, MongoDB
- Spring Security (Authentication & Authorization)
- JWT (JSON Web Token) Implementation
- OAuth2 & Single Sign-On (SSO)
- Role-Based Access Control (RBAC)
- Microservices with Spring Boot 3
- Service Discovery (Eureka)
- API Gateway (Spring Cloud Gateway)
- Circuit Breaker (Resilience4j)
- Distributed Tracing with OpenTelemetry
- Messaging & Event-Driven Architecture
- Kafka & RabbitMQ Integration
- WebSockets for Real-Time Communication
- Spring Boot Caching (Redis, EhCache)
- Asynchronous Processing & Lazy Loading
- Spring Boot Testing (JUnit 5, Mockito, TestContainers)
- Spring Boot Actuator for Monitoring

- Containerization with Docker
- Deploying Spring Boot on Kubernetes
- CI/CD Integration (Jenkins, GitHub Actions)
- GraphQL with Spring Boot
- Serverless with Spring Boot (AWS Lambda, GCP, Azure Functions)
- Logging & Observability (Prometheus, Grafana, ELK Stack)
- Spring Boot 3 & Java 17+ Features
- Performance Optimization & Best Practices

Testing

- Introduction to Software Testing
- Software Development Life Cycle (SDLC)
- Software Testing Life Cycle (STLC)
- Types of Software Testing
 - Functional Testing
 - Non-Functional Testing
 - Manual Testing
 - Automation Testing
- Test Levels
 - Unit Testing
 - Integration Testing
 - System Testing
 - User Acceptance Testing (UAT)
- Test Case Design Techniques
 - Black Box Testing
 - White Box Testing
 - Grey Box Testing
- Test Plan & Test Strategy
- Test Case Writing & Execution
- Defect Life Cycle & Bug Tracking
- Regression Testing & Re-Testing
- Performance Testing
- Load Testing
- Stress Testing
- Scalability Testing

12. Security Testing
13. API Testing (Postman, REST Assured)
14. Automation Testing Tools
 - Selenium
 - Cypress
 - Playwright
 - Appium
15. CI/CD in Testing (Jenkins, GitHub Actions)
16. Agile & DevOps Testing
17. Cloud-Based Testing
18. Mobile Application Testing
19. Database Testing (SQL Queries, Data Validation)
20. Testing Metrics & Reporting

Cloud Deployment Technologies

AWS Cloud Practitioner

- AWS Cloud Concepts & Global Infrastructure
- AWS Shared Responsibility Model
- AWS Compute Services (EC2, Lambda, Elastic Beanstalk)
- AWS Storage Services (S3, EBS, EFS, Glacier)
- AWS Database Services (RDS, DynamoDB, Redshift, Aurora)
- Networking & Content Delivery (VPC, Route 53, CloudFront, ELB)
- AWS Security & Compliance (IAM, Security Groups, KMS, WAF, GuardDuty)
- AWS Pricing Models (On-Demand, Reserved, Spot Instances)
- AWS Cost Management & Billing (Billing Dashboard, Cost Explorer)
- Monitoring & Logging (CloudWatch, CloudTrail, AWS Config)
- Application Integration Services (SNS, SQS, Step Functions)
- AWS Well-Architected Framework (Security, Reliability, Cost Optimization)
- AWS Support Plans & Resources (AWS Whitepapers, Marketplace)

